7.1.1.2 Coil Spring

Its simple layout can be seen in the figure 7.1.1.2. In this type one end of the spring is fixed with the frame, another end is fixed with the lower control arm. When the vehicle rises over a bump the spring gets compressed and it will absorb the energy. When the vehicle gets down from the bump the spring expanded and release the energy.

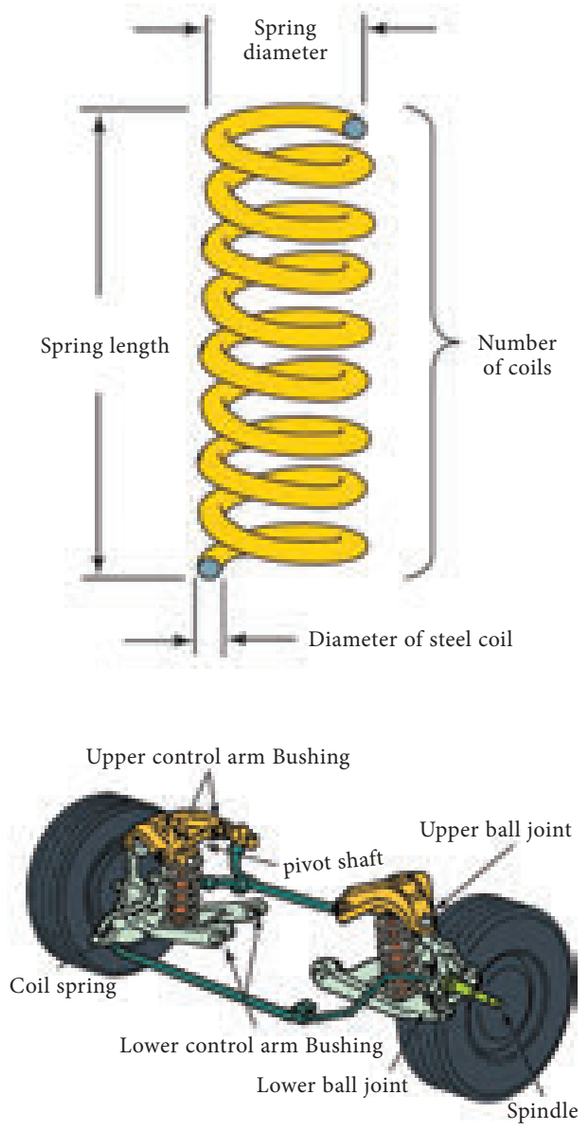


Figure 7.1.1.2 Coil spring

This type of suspension is commonly used in Independent Suspension Systems and in Light Vehicles. It can be used in both front and rear independent suspension systems. It absorbs energy twice than that of leaf spring. Coil spring can withstand shear stress and bending stress.

Advantages

1. Less noise
2. Less weight
3. Less space is enough.
4. Simple in design.

Disadvantages

1. It can not withstand the torque and lateral forces.

7.1.1.3 Stabilizer bar

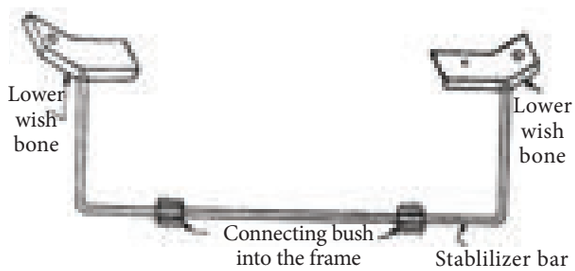


Figure 7.1.1.3 Stabilizer bar

The stabilizer bar is shown in figure 7.1.1.3. The stabilizer bar is mounted in between the two wheels. Stabilizer bars are part of a car's suspension system. They are sometimes also called anti-sway bars or anti-roll bars. The main purpose is to try to keep the car's body from "rolling" in a sharp turn. During turning, body roll occurs when more weight is placed on the outside tires and less weight is placed on the inside tires, which in turn reduces traction and vehicle control. The

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